Muhammad Rayyan Ahmed

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Cover Letter: Internship – Agentic AI and Test-Driven Development

BSH Home Appliances Group Munich 2026-11-05

Dear BSH hiring team,

I am applying for the internship in Agentic AI and Test-Driven Development. Building with large language models and testing that the result really works are the two things my latest project is about.

I am an M.Sc. Data Science student at TU Ilmenau with a B.S. in Artificial Intelligence. I also have industry experience with data models and reporting on SAP platforms.

- **DocuMind,** I built a RAG system over PDF manuals with the Claude API that answered 91% of a 22-question test set correctly and refused all 4 unanswerable questions; hybrid BM25 and semantic retrieval raised MRR from 0.748 to 0.898
- **Testing and CI,** DocuMind has 37 offline unit tests with a fake embedder and a mocked LLM client, GitHub Actions CI and a Docker image
- **Evaluation,** I built an evaluation harness with retrieval metrics and an LLM judge, and used it to trace failures to their cause

I would help build and test LLM-based tools, write tests first where it makes sense and keep CI pipelines green. I have not built AI agents yet; with my RAG and testing experience I can start contributing quickly and learn agent design from your team.

I am willing to relocate to Munich and can start immediately. As a non-EU student I will agree the internship length with you so it fits my permitted working days. I would be glad to discuss the role.

I work in English (IELTS 8.0) and am currently learning German (A1).